

Basic Electronics

Tutorial 4

1. Considering the circuit of Fig. 1, employ superposition to determine the two components of i_8 arising from the action of the two independent sources, respectively.

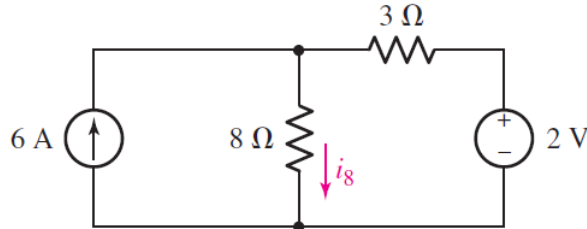


Figure 1: Prob.1

2. (a) Using superposition, determine the voltage labeled v_x in the circuit represented in Fig. 2. (b) To what value should the 2 A source be changed to reduce v_x by 10% ?

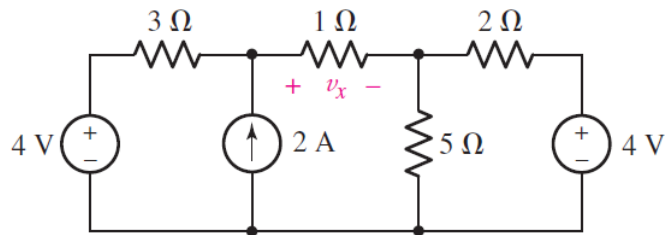


Figure 2: Prob.2

3. (a) Employ superposition to determine the individual contribution from each independent source to the voltage v as labeled in the circuit shown in Fig. 3. (b) Compute the power absorbed by the 2 Ω resistor.

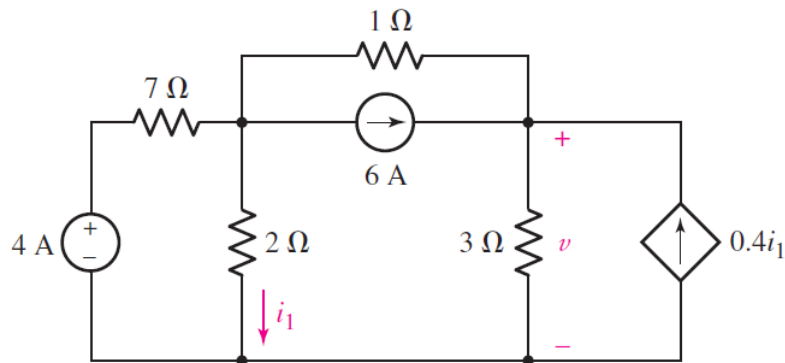


Figure 3: Prob.3

4. For the circuit of Fig. 4, plot i_L versus v_L corresponding to the range of $0 \leq R \leq \infty$.

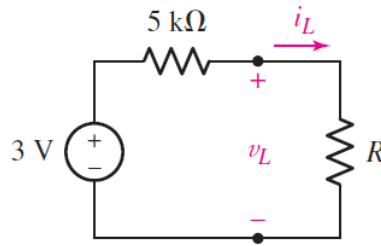


Figure 4: Prob.4

5. Transform the dependent source in Fig. 5 to a voltage source, then calculate V_0 .

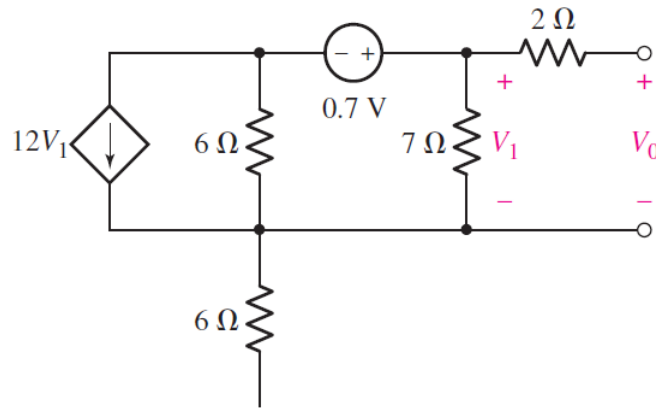


Figure 5: Prob.5

6. With regard to the circuit represented in Fig. 6, first transform both voltage sources to current sources, reduce the number of elements as much as possible, and determine the voltage v_3 .

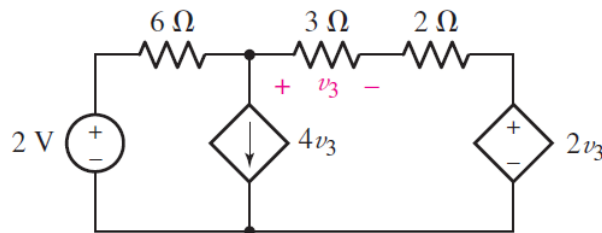


Figure 6: Prob.6

7. Determine the Thévenin equivalent of the network shown in Fig. 7 as seen looking into the two open terminals.
8. (a) Obtain a value for the Thévenin equivalent resistance seen looking into the open terminals of the circuit in Fig. 8 by first finding V_{oc} and I_{sc} . (b) Connect a 1 A test source to the open terminals of the

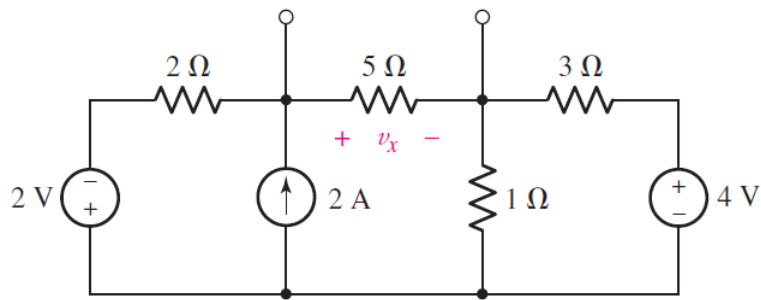


Figure 7: Prob.7

original circuit after shorting the voltage source, and use this to obtain R_{TH} . (c) Connect a 1 V test source to the open terminals of the original circuit after again zeroing the 2 V source, and use this now to obtain R_{TH} .

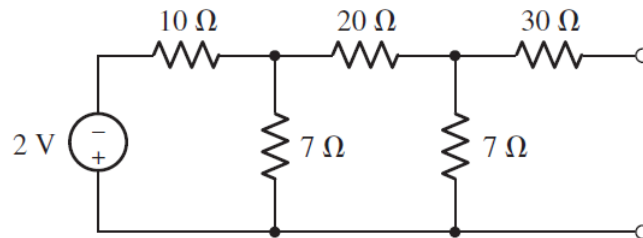


Figure 8: Prob.8

9. Determine the Thévenin and Norton equivalents of the circuit represented in Fig. 9 from the perspective of the open terminals. (There should be no dependent sources in your answer.)

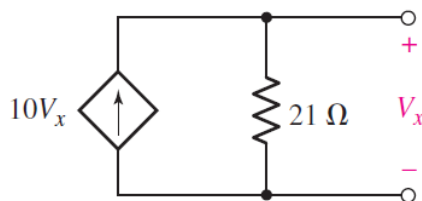


Figure 9: Prob.9

10. Determine the Norton equivalent of the circuit drawn in Fig. 10 as seen by terminals a and b. (There should be no dependent sources in your answer.)

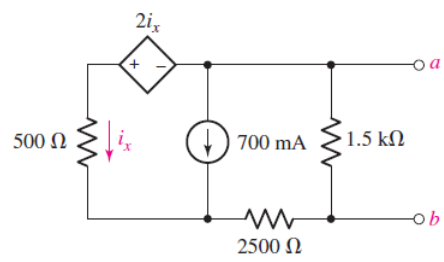


Figure 10: Prob.10